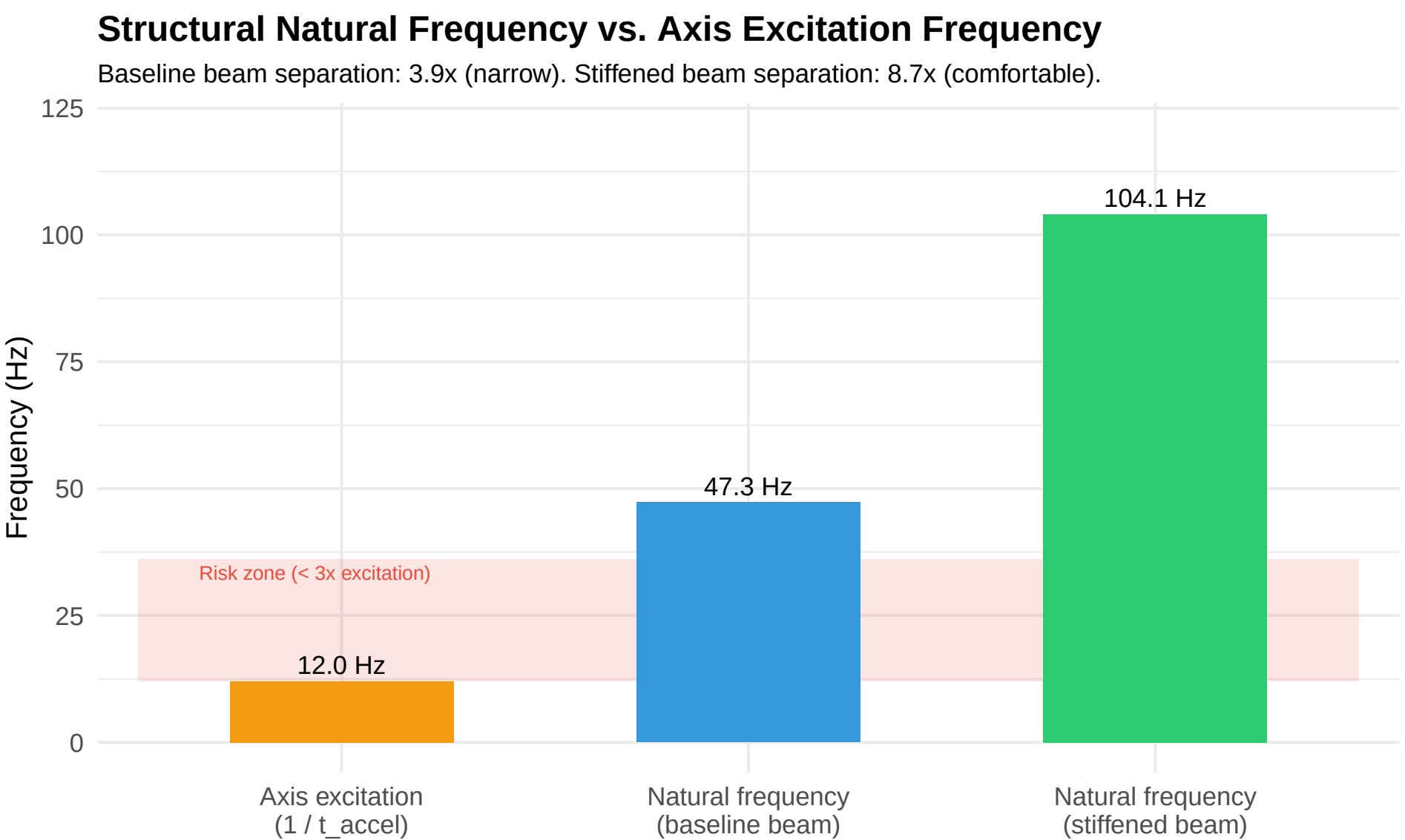
Structural Natural Frequency vs. Axis Excitation Frequency
Baseline beam separation: 3.9x (narrow). Stiffened beam separation: 8.7x (comfortable).
Frequency (Hz)
125
100
75
50
25
0
Risk zone (< 3x excitation)
12.0 Hz
47.3 Hz
104.1 Hz
Axis excitation
(1 / t_accel)
Natural frequency
(baseline beam)
Natural frequency
(stiffened beam)